Mason Rosenau

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Experience

Software Engineer Nov. 2025 – Present
Garmin *Salem, OR*

Software Engineering Intern June 2025 – Nov. 2025
Garmin *Salem, OR*

- Implemented features in C in accordance with requirements for a **multi-threaded, embedded** avionics project
- Authored formal software requirements along with **automated, class-based Python tests** and low-level C tests to ensure software compliance with aviation safety standards like DO-178B
- Collaborated with multiple teams during software release cycles to ensure successful integration with other units

DevOps Engineering Intern Apr. 2024 – Sept. 2024
Intel Corporation *Hillsboro, OR*

- Reduced the time to deploy and update infrastructure on regression testing platforms with Python automation
- Removed the use of **13** secrets in client-server architecture codebases using environment variables and SFTP
- Setup and automated MySQL database transfers to ensure a more **up-to-date** development environment
- Implemented in-browser binary downloads, reducing the previous workflow time by **over 90 percent**
- Implemented the initialization of multiple registers in **C firmware** according to design team specifications
- Coordinated with lab technicians to help setup test platforms with the next generation of silicon

Education

Oregon State University Sept. 2021 – June 2025
Bachelor of Science in Computer Science - GPA: 4.0 *Corvallis, OR*

- **Coursework:** Computer Architecture, Operating Systems II, Computer Organization and Assembly Language, Parallel Programming, Translators, Mobile Software Development, Computer Graphics, Analysis of Algorithms
- **Awards:** Drucilla Shepard Smith Scholastic Awardee (3x), College of Engineering Dean's List

Projects

Hobby Websites | *Django, Cloudlfare, HTML, CSS, JavaScript, FastHTML* | May 2023 – Present

- Various websites I've created including little notes for my partner, a website for our wedding and registry, and various math games and tools, all hosted on my Raspberry Pi

HempDB (Senior Capstone) | *Django, LeafletJS, Redis* | Oct. 2024 – June 2025

- Lead a team of 6 developers as **project manager** working on OSU's Industrial Hemp Stakeholder Database
- Implemented a map of **over 5,000 companies** featuring heatmaps, filtering, caching, and automatic geocoding
- Created an administrator email notification system to expedite the approval of pending changes

Chart Builder Test Suite | *Cypress, GitHub Actions, Jest* | June 2023

- Wrote end-to-end tests for a web application by **simulating user interaction** using Cypress
- Constructed a **CI/CD pipeline** using GitHub Actions to ensure an **errorless build and deployment** process
- Focused on a **user-centered** testing philosophy while collaborating with teammates using GitHub

OTP Encryption & Decryption | *C, Sockets, Ports, Key Generation* | May 2023 – June 2023

- Built a client-server application that can concurrently encrypt and decrypt text using one-time pad encryption
- Learned more about client-server relationships, concurrency, and the importance of verifying connections

Small Shell | *C, UNIX* | Apr. 2023 – May 2023

- Developed a shell to handle UNIX **command line** functionalities using signal handling, file I/O, and C strings
- Learned complex program flow and the importance of memory and process management

Skills & Interests

Languages: Python, C, C++, Kotlin, Assembly (x86, MIPS, AVR), HTML, JavaScript, Bash
Tools: pytest, Visual Studio, Docker, GitHub Actions, Cloudflare, Vercel, Django, Flask, Redis, MySQL, SQLite, GCP
Personal Interests: Gaming, MMA, Tennis, Freestyle Drone Flying, Video Creation, Strongman, Photography